

Central Bank Dissonance and Asset Returns

A Structural Decomposition Approach to Monetary Regime Detection

Thesis Proposal & Working Plan · Draft v0.1

Sample. A panel of fourteen economies, comprising seven developed and seven emerging markets, observed monthly over 2005–2024.

Research question. Do measures of internal inconsistency within and across the components of central bank policy contain predictive information for asset returns beyond established regime-change detectors and standard risk factors?

Contribution. A unified framework for measuring central bank internal consistency, instantiated empirically across a DM/EM panel and benchmarked against statistical, market-based, and macro early-warning regime detectors.

Pathway. Master's deliverable with explicit doctoral upgrade option; same architecture supports both.

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Part I — Thesis Proposal

1. Background and Motivation

Central bank policy is a dominant determinant of asset prices across both developed and emerging market economies, yet its empirical characterisation in the literature remains fragmented. Taylor-rule estimation (Clarida-Gali-Gertler 1999; Aron-Muellbauer 2007) treats the policy rate as the bank's coherent response to inflation and output. The forward guidance literature (Bauer-Swanson 2023; Campbell-Evans-Fisher-Justiniano 2012; Gürkaynak-Sack-Swanson 2005) isolates communication as a separate transmission channel. Macroprudential studies (Cerutti-Claessens-Laeven 2017) examine financial-stability tools in isolation from monetary policy. The intermediary-frictions literature (He-Krishnamurthy 2013; Adrian-Etula-Muir 2014) considers how policy is filtered through banks, dealers, and non-bank financial institutions. Each of these traditions has produced robust results within its frame. But each studies a single component of what is operationally a unified decision-making apparatus deploying multiple instruments through multiple transmission channels under multiple constraints.

The empirical gap is not in studying any one component more carefully. It is in the *integration* — specifically, in what happens when the components are internally inconsistent.

The 2013 Federal Reserve "taper tantrum" provides one canonical case. On 22 May 2013, Chairman Bernanke testified that the Federal Open Market Committee could begin reducing the pace of asset purchases later that year. The federal funds target was unchanged. The balance sheet was unchanged. Only the forward guidance on the path of asset purchases shifted. Yet ten-year Treasury yields rose from 1.93% to over 3.0% by September; emerging market currencies, bonds, and equities sold off sharply; and the so-called Fragile Five — Brazil, India, Indonesia, South Africa, and Turkey — experienced severe capital flow reversals. Investors reading only the policy rate observed unchanged monetary stance. Investors reading the full instrument vector — including the forward guidance path factor identified in the GSS-style identification literature — observed sharply tighter monetary conditions. The headline summary "Fed on hold" was a misleading scalar reduction of an instrument vector whose components were not coincident.

A second canonical case is the European Central Bank's 2011 cycle. In April and again in July 2011, the ECB raised the main refinancing rate by 25 basis points, citing inflation concerns. These hikes occurred during a deepening sovereign debt crisis in which peripheral eurozone banks held large positions in distressed sovereign paper. The standard reaction function — inflation above target — implied tightening. The institutional state of the banking system — capital strained, sovereign exposures concentrated, market funding deteriorating — implied loosening. The components of policy were misaligned. Following Mario Draghi's accession in November 2011, the hikes were reversed.

A third recent case is the 2023 collapse of Silicon Valley Bank and the subsequent regional banking stress in the United States. The Federal Reserve was continuing an aggressive hiking cycle (the L3 monetary instrument tightening); simultaneously, the asset-liability mismatch on bank balance sheets (an L5 intermediary state) was rupturing under that same tightening. The system was signalling a dissonance between the instrument deployed and the intermediary capacity to absorb it. The Bank Term Funding Programme — a new L3 liquidity facility — was created in response.

In each case, the central bank system was internally inconsistent before market-based regime indicators registered the shift. This thesis terms such inconsistency *dissonance* and investigates whether systematic measurement of dissonance, derived from a structural decomposition of central bank policy, contains predictive content for asset returns.

2. Research Question

The thesis investigates a single question:

Do measures of internal inconsistency within and across the components of central bank policy — the reaction function, the loss function and its binding constraints, the instrument vector, communication, transmission to intermediaries, and the expectations and financial conditions they generate — contain predictive information for asset returns beyond that contained in established regime-change detectors and standard risk factors?

The investigation is conducted strictly through the lens of monetary policy and the central bank. It is not a general macroeconomic regime-change detection exercise.

3. Theoretical Framework

The framework is built on three primitives.

The structural object.

The central bank is represented as a directed graph $G = (N, E)$, where N is a set of nodes partitioned across ten hierarchical levels — strategic core (reaction function), optimisation primitives (loss function and constraints), instruments, markets, intermediaries, expectations, financial conditions, valuation drivers, asset prices, and terminal outputs — and E is a set of directed edges representing transmission and feedback relationships. The hierarchy is operational but not strict; feedback edges from terminal nodes to the reaction function close the system.

Node taxonomy.

Each node belongs to one of three species: *agent nodes* (the central bank itself, banks, NBFIs, dealers, households/corporates, sovereign — economic actors with balance sheets and behavioural rules); *market nodes* (yield curve, FX, inflation-linked bonds, vol surface — price systems governed by no-arbitrage); and *instrument nodes* (policy rate, forward guidance, balance sheet, FX intervention, macroprudential, liquidity facilities — actions deployed by agents that operate on markets and intermediaries, transmitting subject to a state-dependent multiplier).

The consistency model.

For any node n with components $\{c_1, \dots, c_k\}$, a consistency model R_n maps the components to zero in the equilibrium regime: $E[R_n(c_1, \dots, c_k)] = 0$. Dissonance at node n at time t is defined as $\text{Dissonance}_n(t) = |R_n(c_1(t), \dots, c_k(t))|$. Node n is in normal regime if $\text{Dissonance}_n \leq \tau_n$, where τ_n is a country-specific threshold.

Dissonance taxonomy.

Four structural types are distinguished: *intra-node* (components within a single node misalign — e.g., Taylor residual within L1); *inter-node* (peer nodes pull apart — e.g., monetary easing with macroprudential tightening within L3); *vertical* (cross-level misalignment — e.g., instruments tighten while markets ease); and *temporal* (a node behaves out of regime relative to its own history — e.g., Taylor coefficient drift).

Aggregation.

The country-level composite is $D_{c,t} = \sum_n w_n \times \text{Dissonance}_{n,c}(t)$, with four candidate weighting schemes (equal-weight, hierarchical, PCA, theory-weighted). Robustness of $D_{c,t}$ to weighting choice is treated as a result.

Architecture universality and parameter specificity.

The graph G is country-invariant; country specificity enters through which sub-nodes are populated, numerical parameter values, and threshold non-linearities. A complete formal statement of the framework — including definitions, edge types, conditional transmission multipliers, constraint binding, and the system regime characterisation — is contained in Chapter 2 of the planned thesis.

4. Methodology and Empirical Strategy

Sample.

A panel of fourteen economies, comprising seven developed markets and seven emerging markets, observed monthly over the period 2005–2024:

- *Developed markets*: United States, United Kingdom, Eurozone (treating the ECB as the single relevant monetary authority), Japan, Canada, Australia, Sweden.
- *Emerging markets*: Brazil, Mexico, South Africa, Indonesia, India, Chile, Turkey.

The DM/EM split is methodologically important. It permits direct comparison of dissonance properties across institutional regimes with very different constraint sets, credibility levels, and intermediary structures. It also addresses a likely examiner question: does the framework generalise beyond emerging markets? The contrast between DM and EM dissonance is treated as a substantive contribution rather than a robustness check.

Estimation.

For each node and each country, a consistency model R_n is estimated using established methods from the relevant literature: country-specific Taylor rules (Aron-Muellbauer for SA, Mohanty-Klau for EM panel, CGG-style for DM); macroprudential action indices (IMF iMaPP database, Alam et al. 2019); forward guidance path factors (GSS/Bauer-Swanson identification where OIS markets permit; NLP-based sentiment analysis on policy statements elsewhere); intermediary stress composites (BIS Financial Soundness Indicators); expectations gaps (survey vs market-implied); and term premium decompositions (Adrian-Crump-Moench).

Aggregation.

The composite $D_{c,t}$ is constructed under each of the four weighting schemes specified in Section 7 of the framework, with the equal-weight scheme as baseline.

Predictive testing.

$D_{c,t}$ is tested as a predictor of asset returns at multiple horizons (1, 3, 6, 12 months) across asset classes: local-currency sovereign bonds at standard maturities, exchange rates, equity indices, and sovereign credit default swaps. Both within-country regressions and panel regressions with country and time fixed effects are reported. Out-of-sample testing uses an expanding-window scheme with the first cut at 2015.

Benchmarks.

The dissonance composite is tested against three classes of established baselines: statistical regime filters (Hamilton 1989 two-state Markov-switching; Bai-Perron break tests on reaction functions; Primiceri 2005 TVP-VAR); market-based stress indicators (FCI z-scores; Gilchrist-Zakrajšek 2012 excess bond premium analogue; sovereign CDS thresholds); and macro early-warning systems (Kaminsky-Reinhart 1999 signals approach; Schularick-Taylor 2012 credit indicators). Encompassing regressions test for independent information content after these controls.

Robustness.

The empirical strategy includes drop-one-node analyses (which results survive removing each node), drop-one-country analyses, alternative aggregation schemes, alternative threshold τ_n calibrations, and split-sample tests across crisis and non-crisis subsamples.

5. Expected Contributions

The thesis makes three contributions to the existing literature.

Theoretical.

It develops a unified framework for measuring internal consistency within a central bank's decomposed structure. The dissonance metric R_n , applied at each node of the decomposition, formalises an idea that practitioners use intuitively but that academic monetary economics has not systematically operationalised — that the policy stance is a vector, not a scalar, and that meaningful information resides in the consistency relations among its components. The framework bridges otherwise-separate literatures on reaction-function estimation, forward guidance identification, macroprudential policy, and intermediary frictions through a common structural object.

Empirical.

The thesis tests the dissonance framework as a predictor of asset returns across a fourteen-country DM/EM panel, against three classes of established baselines. The hypothesis is not that dissonance dominates these methods uniformly, but that it contains *independent* information after controls and provides *attribution* — identifying which transmission node broke rather than only that a regime shift occurred. The DM/EM comparison is itself an empirical contribution: documenting whether dissonance properties (frequency, magnitude, predictive horizon, asset-class incidence) differ systematically across institutional regimes.

Methodological.

The framework is constructed as a regime-change detector grounded in transmission theory rather than fitted to historical episode features. This addresses the documented out-of-sample fragility of reduced-form crisis indicators (Berg-Pattillo 1999; Frankel-Saravelos 2012), which have repeatedly performed worse on episodes outside their calibration sample. By deriving signals from structural decomposition rather than empirical pattern-matching, the framework is intended to generalise more cleanly across regimes and time periods.

6. Possible Findings

Three outcomes are pre-specified as informative.

Positive: dissonance signals predict asset returns with independent information content after controls; the strongest contribution comes from cross-instrument and cross-level dissonance; the signal leads established market-based indicators by economically meaningful horizons; magnitudes are larger in EM than DM but the framework operates in both.

Null: the signals reproduce known factors and offer no additional information.

Conditional (theoretically most likely): dissonance predicts only at regime transitions and not in calm periods, consistent with its theoretical character as a regime-change detector with episodic rather than continuous information content. Unconditional Sharpe is modest; conditional Sharpe around fired signals is strong.

The thesis is constructed to be publishable under any of these outcomes.

7. Indicative Timeline

The plan spans approximately 24 months of part-time work alongside full-time employment at an asset manager. The structure is sequential: framework lock before empirics, pilot country before panel, predictive tests before mechanism analysis.

- **Month 1:** proposal finalisation; supervisor engagement; data access confirmation.
- **Months 2–4:** literature review across six strands; theoretical framework lock; pre-registration of node-level specifications, thresholds, and weighting schemes.
- **Months 5–8:** data infrastructure construction across all fourteen countries and all nodes; documentation of every data source, transformation, and proxy decision.
- **Months 9–12:** South Africa pilot implementation of the full framework end-to-end; sanity checks against known regime events; preliminary predictive analysis.
- **Months 13–16:** panel expansion to remaining thirteen countries; cross-country harmonisation; pooled descriptive analysis.
- **Months 17–20:** predictive testing across asset classes and horizons; encompassing regressions against baselines; robustness analysis.
- **Months 21–22:** mechanism analysis and case studies; node-level attribution; conditional Sharpe analysis.
- **Months 23–24:** final thesis drafting; revision rounds; defence preparation.

Pre-registration of the framework specification at the conclusion of Month 4 is the central credibility-protecting discipline of the project.

8. Doctoral Upgrade Pathway

The proposal is structured to permit upgrade to doctoral candidature subject to quality of empirical results and supervisor recommendation. The same architecture supports both outcomes: the master's deliverable instantiates the framework with baseline specifications; the doctoral extension adds regime-switching estimation at each node (Markov-switching Taylor rules; time-varying loss-function weights), structural interpretation of dissonance episodes as identified monetary shocks, additional robustness across alternative country sets and time windows, and deeper development of cross-level dissonance — which is the theoretically most novel component of the framework. The upgrade decision is informed by Phase 5 results, not pre-committed.

9. Sample Caveat

A panel of fourteen countries is at the upper end of what is feasible at the master's level given the per-node data construction burden across forty-eight estimated R_n 's. If data quality constraints render full instantiation across the panel infeasible in the available timeframe, the sample will be reduced — most likely by trimming to five DM and five EM economies, retaining the comparative structure but reducing per-country implementation cost. The decision is taken at the end of Phase 2, based on documented data availability, and recorded in the data appendix.

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Part II — Working Plan

The plan is structured in eight phases over approximately 24 months, assuming part-time work alongside full-time employment at an asset manager — roughly 15–20 hours per week, weighted toward weekends. The doctoral upgrade extends Phases 4–7 with depth at every node, but the architecture is the same.

The principle throughout: **do nothing in parallel that can be done sequentially**. Build framework, then empirics. Build pilot, then panel. The thesis is large enough that compounding errors from premature parallelisation cost more than sequential discipline saves.

Phase 0 — Pre-flight (weeks 1–4)

Goal: Convert the framework into a written proposal and secure institutional commitment.

Tasks:

- Draft 8–10 page proposal: motivation, research question, framework (the methodology chapter), planned empirical design, scope, contribution, expected outcomes, indicative timeline.
- Identify 2–3 potential supervisors. Criteria: published in EM monetary economics or empirical asset pricing; comfortable with ambitious-scope theses; openness to doctoral upgrade path.
- Initial supervisor meetings: present proposal, gauge fit, get feedback.
- Lock supervisor and degree pathway — explicitly agree the doctoral upgrade option upfront.
- Confirm Bloomberg/Refinitiv access through Ninety One for thesis use (data licensing agreement may be needed).

Deliverable: signed supervisor letter, approved proposal, data access cleared.

Decision point: supervisor alignment with the doctoral upgrade path. If supervisor only supports a self-contained master's, scope must be revisited before Phase 1.

Phase 1 — Literature & Framework Lock (months 2–4)

Goal: Comprehensive literature review and locked theoretical framework — Chapter 2 of the eventual thesis.

Tasks:

Literature review across six strands:

- Reaction functions: Taylor 1993, Clarida-Galí-Gertler 1999/2000, Aron-Muellbauer 2007, Mohanty-Klau.
- Forward guidance identification: Gürkaynak-Sack-Swanson 2005, Campbell-Evans-Fisher-Justiniano 2012, Bauer-Swanson 2023, Nakamura-Steinsson 2018.
- Macroprudential policy: Cerutti-Claessens-Laeven 2017, Alam et al. (iMaPP), Akinci-Olmstead-Rumsey.
- Intermediary frictions: He-Krishnamurthy 2013, Adrian-Etula-Muir 2014, Du-Tepper-Verdelhan on CIP.
- Regime-switching macro: Hamilton 1989, Bianchi 2013, Davig-Leeper, Bai-Perron, Primiceri 2005.
- EM monetary economics: Calvo-Reinhart, Rey GFC, Mendoza, Gourinchas.
- Finalise framework chapter using the methodology draft as spine; refine with literature anchors.
- **Pre-register the framework spec:** nodes, edges, R_n specifications, weighting schemes, τ_n thresholds. Timestamped and version-controlled — this is the single most important

credibility-protecting document of the project.

- Confirm fourteen-country panel and justify selections explicitly (data availability, institutional variation, dissonance episode frequency).

Deliverable: literature review document (~30 pages), finalised methodology chapter, pre-registration document.

Decision point: Is the framework tight enough to instantiate empirically? If R_n for any node is still ambiguous, return to spec.

Phase 2 — Data Infrastructure (months 5–8)

Goal: Build the database. This is consistently the most underestimated phase. Budget 30% more time than your first estimate.

Tasks by node:

- **L1:** country policy rates, CPI, output gap proxies, FX REER, Fed funds rate. Sources: BIS, IMF IFS, central bank websites.
- **L2:** debt-to-GDP, fiscal deficit, FX reserves, ELB indicators. Sources: IMF WEO, central banks.
- **L3:** rate decisions, balance sheet sizes, FX intervention metadata, macropru actions (IMF iMaPP), communication transcripts.
- **L4:** yield curves at standard maturities, FX spot/forward, BEI curves, vol surfaces. Source: Bloomberg.
- **L5:** BIS Financial Soundness Indicators (banks); money market spreads (NBFI proxy); CIP basis (dealers); BIS DSR (HHs/corps); sovereign CDS and rating changes.
- **L6:** survey forecasts (Focus Brazil, Banxico, Reuters polls, Consensus Economics).
- **L7:** country-specific FCIs where available; construct from components where not.
- **L8/L9:** local-currency bond returns at standard maturities, FX returns, equity returns, sovereign credit returns.

Cross-cutting tasks:

- Database design: PostgreSQL (or similar); standardised schema per node.
- Country-time-frequency harmonisation.
- Documentation of every source, transformation, and proxy decision — this becomes the Data Appendix.
- Quality flagging per node per country.
- Version control.

Deliverable: complete panel database with documentation; data appendix first draft.

Decision point: Are there nodes for which data is too thin to instantiate R_n ? Document explicitly as scope limitation; do not improvise proxies without justification.

Phase 3 — South Africa Pilot (months 9–12)

Goal: Implement the full framework on one country end-to-end before scaling. South Africa chosen given Ninety One position and personal familiarity.

Tasks:

- Estimate R_n for every node on SA 2005–2024.

- Compute Dissonance $D_{n,SA}(t)$ for every node at monthly frequency.
- Compute composite $D_{SA,t}$ under all four weighting schemes.
- Plot dissonance time series; identify SA regime events (2008–09 GFC, 2013 taper, 2015–16 commodity crash, 2018 EM selloff, 2020 COVID, 2022 hiking cycle, fiscal stress episodes).
- Sanity check: does the composite fire around known SA regime events with reasonable timing?
- Initial predictive regression: $D_{SA,t}$ on SA asset returns at 1/3/6/12-month horizons.
- Document every implementation choice not specified in the pre-registration.

Deliverable: working SA pipeline; descriptive analysis of SA dissonance 2005–2024; preliminary predictive results.

Decision point: Does the framework produce sensible output on SA? If composite fires constantly, never, or at random times, the framework needs revision *before* scaling. Better to find this now than after building all fourteen countries.

Phase 4 — Panel Expansion (months 13–16)

Goal: Replicate SA pipeline across the remaining thirteen countries.

Tasks:

- Implement R_n for each country, respecting country-specific parameterisation.
- Country-by-country sanity checks against known regime events.
- Cross-country harmonisation of frequencies and scales.
- Pooled panel construction with country fixed effects.
- Identify country-specific node coverage gaps; document them.

Deliverable: complete panel of country-time dissonance time series; cross-country descriptive plots.

Phase 5 — Predictive Testing (months 17–20)

Goal: Test the central hypothesis — does $D_{c,t}$ predict asset returns beyond established baselines?

Tasks:

- Within-country predictive regressions of asset returns on $D_{c,t}$ at multiple horizons (1, 3, 6, 12 months).
- Panel regressions with country and time fixed effects.
- Out-of-sample testing: train pre-2015, test 2015–2024 (or expanding-window scheme).
- Encompassing regressions adding Hamilton MS regime probability, FCI z-score, Gilchrist-Zakrajšek excess bond premium analogue, Kaminsky-Reinhart-style signal.
- Lead-lag analysis vs each baseline.
- AUC/ROC for regime-shift prediction at 3/6/12-month leads.
- Robustness across the four weighting schemes.
- Drop-one-node robustness: which results survive removing each node?

Deliverable: full empirical results — tables, figures, robustness analysis.

Decision point: at this stage you know whether the outcome is positive, null, or conditional. The thesis narrative branches accordingly.

Phase 6 — Mechanism and Case Studies (months 21–22)

Goal: Explain *why* the composite signal predicts (or doesn't), and demonstrate attribution.

Tasks:

- Case studies: 2013 Fed taper, ECB 2011 cycle, SVB 2023, Turkey 2018, Argentina 2023, SARB 2020 and 2022–23 cycles. For each: plot composite and node-level decomposition around the event. Show which nodes fired and when.
- Conditional Sharpe analysis: signal performance in transition regimes vs equilibrium regimes (validates the regime-change-detector framing).
- Node-level attribution: which nodes carry the most weight in the composite? Which carry the most predictive power?
- Asset-level attribution: when D fires, can the node-level decomposition predict *which* sub-asset responds most strongly?

Deliverable: mechanism chapter with case studies and attribution analysis.

Phase 7 — Writing & Defence (months 23–24)

Tasks:

- Final thesis draft assembly (Chapter 1 motivation, Chapter 2 framework, Chapter 3 empirical strategy, Chapter 4 SA pilot, Chapter 5 panel results, Chapter 6 mechanism, Chapter 7 conclusion).
- Internal revision rounds with supervisor.
- External examiner identification (especially relevant if pursuing doctoral upgrade).
- Defence preparation.
- **Doctoral upgrade decision:** based on quality of empirical results, supervisor recommendation, and your own appetite. The decision is informed, not pre-committed.

Deliverable: submitted thesis.

Cross-cutting Discipline Rules

- **Pre-registration is non-negotiable.** Every R_n , every τ_n , every weighting scheme timestamped before empirical work runs. Modifications during empirics documented and justified, never silent.
- **Version-control everything.** Git for code, separate repo for the thesis manuscript, separate folder for pre-registration with file hashes.
- **Sequential, not parallel.** The temptation to start the panel before SA pilot works is the dominant risk to the timeline.
- **Documentation as you go.** Data appendix written during Phase 2, not at the end. Code commented inline. Decisions logged in a research diary with timestamps — this becomes the answer to "why did you do X?" in the defence.
- **Allocate slack.** Add 30% to every time estimate. Data construction in particular is consistently underestimated.
- **Standing supervisor slot.** One conversation per month minimum, monthly written progress note. Surprises in month 18 are project-ending; surprises in month 3 are recoverable.
- **Reproducibility from day one.** Code that can re-run the entire pipeline end-to-end. If a result cannot be regenerated by running one script, it does not go in the thesis.
- **Honest scoping in real time.** When a node turns out to be unmeasurable in a country, document it and move on. The temptation to improvise proxies is what turns rigorous theses into critiqued ones.

The first concrete action item is Phase 0, Task 1: draft the proposal document into a form to circulate to potential supervisors. The framework chapter serves as the spine; surrounding it with motivation, empirical design summary, and timeline produces the proposal. Two weekends of focused work.